

Unemployment: Causes and Consequences

ECON 300 - Labour Economics

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Learning Objectives (1/2)

- Explain how imperfect information can generate the coexistence of unemployment and vacancies.
- Assess how technological change, especially the Internet, affects job search and matching.
- Show how demand shifts and sectoral change can generate unemployment.
- Summarize evidence on job displacement and its consequences.

Learning Objectives (2/2)

- Describe models where firms pay above-market wages: efficiency wages, implicit contracts, and insider–outsider models.
- Explain the rationale for unemployment insurance/employment insurance and its effects on labour supply and unemployment.
- Summarize empirical evidence on UI/EI, incidence, and duration of unemployment.

- What role does wage rigidity play in unemployment?
- How does imperfect information create frictional unemployment?
- Is online search improving matching efficiency?
- What are the consequences of job loss for workers and families?
- Does UI/EI do more harm than good?

Causes of Unemployment

- Minimum wage laws
- Institutional factors such as unions or a large public sector
- Demand factors and seasonal unemployment
- Frictional unemployment arising from turnover and imperfect information
- Sectoral shifts causing structural unemployment
- Generous UI/EI affecting search incentives

Definitions

- **Frictional unemployment:** short-term unemployment arising from normal job search and turnover.
- **Structural unemployment:** unemployment caused by mismatches between worker skills/locations and available jobs.
- **Seasonal unemployment:** recurring unemployment related to seasonal patterns in industries such as tourism, agriculture, or construction.

Interpreting the Different Causes

- Not all unemployment has the same cause, so not all unemployment should be addressed with the same policy.
- Frictional unemployment may require better information and matching.
- Structural unemployment may require retraining or geographic mobility.
- Cyclical unemployment may require macroeconomic stimulus.

Real Example

A laid-off oil worker in Alberta may face structural unemployment if jobs are growing mainly in healthcare or technology rather than energy.

Search Unemployment: Core Ideas

- Imperfect information delays matching between workers and firms.
- Searching for jobs has both costs and benefits.
- Workers adopt a **stopping rule** or **reservation criterion**: accept the first offer that meets a minimum acceptable standard.
- Optimal search balances expected marginal benefit and expected marginal cost.

Why Search Takes Time

- Workers do not observe all vacancies instantly.
- Firms do not immediately identify the best applicants.
- Search time can improve job match quality, but it also delays income.

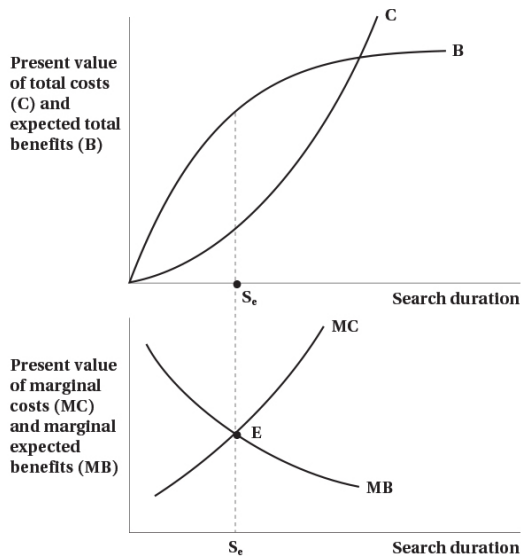
Interpretation

A longer search may be rational if it increases the chance of finding a better job.
Unemployment is not always wasteful; some of it is part of efficient job matching.

Real Example

A recent graduate may reject a low-wage unrelated job and search longer for an entry-level job in their field because the long-run payoff is better.

Figure 17.1: Optimal Job Search



Reading Figure 17.1

- In the top panel, total expected benefits of search rise at first but eventually flatten.
- Total costs rise as search continues.
- In the bottom panel, optimal search occurs where **marginal expected benefit equals marginal cost**.

Interpretation

The optimal search duration is not necessarily zero. It is efficient to keep searching until the extra expected gain from one more unit of search just equals its cost.

Worked Example: Reservation Wage and Hazard

Julia faces three job offers with equal probability: \$2,400, \$2,200, and \$2,000 per month.

- Expected value of the next offer:

$$E[W] = \frac{2400 + 2200 + 2000}{3} = 2200$$

- Reservation wage:

$$W^R = 2200$$

- She rejects any offer below \$2,200 and accepts the first offer at or above \$2,200.
- Acceptance probability:

$$P(W \geq 2200) = \frac{2}{3}$$

Definition

The **hazard rate** is the probability that an unemployed worker exits unemployment in a given period, conditional on still being unemployed.

Interpreting Reservation Wage

- Higher search cost lowers the reservation wage.
- Better future offer prospects raise the reservation wage.
- Higher unemployment benefits may allow workers to search longer and hold out for better jobs.

Real Example

A worker with savings or EI benefits may reject a very low offer and continue searching, while a worker with no buffer may accept quickly.

Imperfect Information: Market Implications

- Markets do not clear instantaneously.
- Wage dispersion can persist even with homogeneous workers and jobs.
- Individual firms may face upward-sloping labour supply because search frictions reduce worker mobility.

Definition

Dynamic monopsony refers to a setting where individual firms have some wage-setting power because workers face frictions in moving between jobs.

Why Wage Dispersion Can Persist

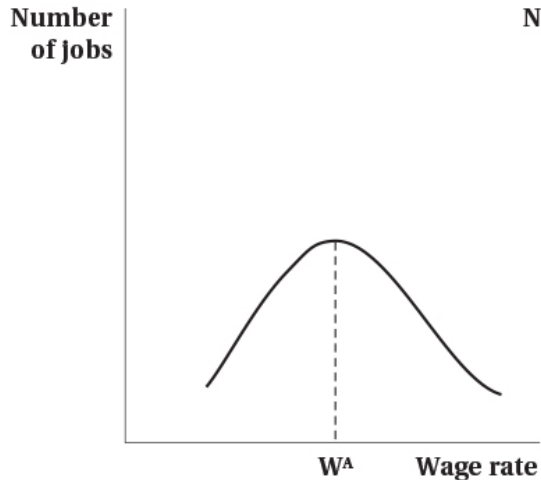
- If workers do not observe all wages, some firms can pay less and still attract workers.
- Search frictions prevent immediate movement to the best-paying employer.
- This helps explain why similar workers may earn different wages in similar jobs.

Real Example

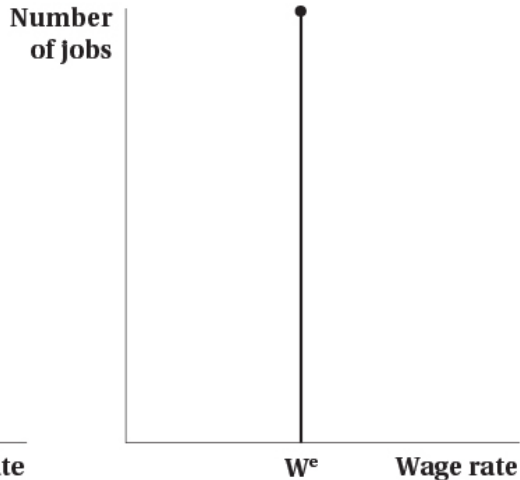
Two retail firms in the same city may pay different wages because workers do not instantly know all job offers and switching jobs is costly.

Figure 17.2: Wage Distributions Under Imperfect vs. Perfect Information

(a) Imperfect information



(b) Full information



Reading Figure 17.2

- Under imperfect information, wages are spread over a distribution.
- Under full information, wages collapse to a single market-clearing wage.

Interpretation

Search frictions help explain both unemployment and persistent wage differences. Without frictions, workers would all move immediately to the highest available wage and wage dispersion would disappear.

Search, the Internet, and Matching

- Online job boards, employer portals, and screening tools have grown rapidly.
- Technology can raise contact rates between workers and firms.
- Better screening may improve match quality and shorten vacancy duration.

Interpretation

Technology may reduce frictional unemployment by lowering search costs, but it can also raise application volume and screening complexity for employers.

Has Online Search Solved Search Frictions?

- It often makes vacancy information easier to access.
- But search is still imperfect:
 - workers may not know which applications are worth sending,
 - firms may struggle to process many applications,
 - online systems may filter out qualified candidates.

Real Example

A worker can apply to 200 jobs online in a week, but still remain unemployed if applications are poorly targeted or automated filters reject them.

Sectoral Shifts and Unemployment

- Major structural change raises unemployment during reallocation.
- Highly cyclical industries amplify unemployment in downturns.
- Regional reallocation in Canada drives migration, retraining, and search.

Definition

Sectoral shifts are changes in the pattern of labour demand across industries or regions.

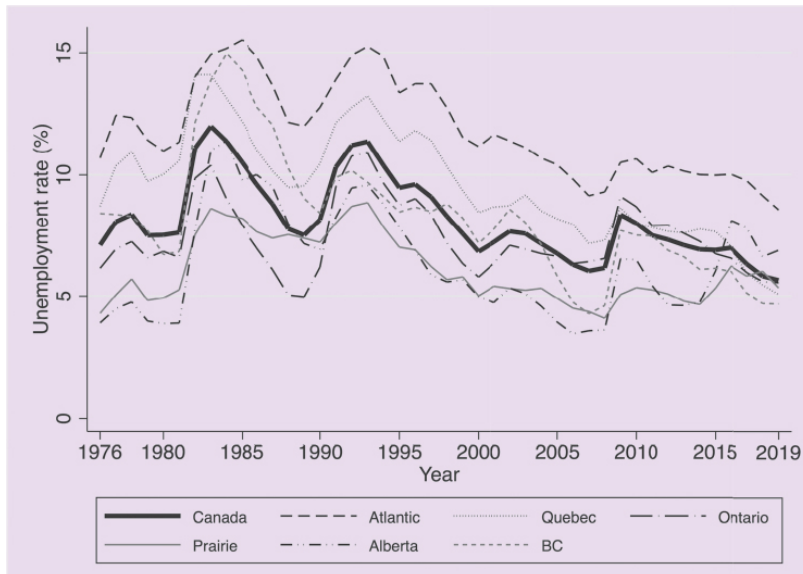
Why Sectoral Shifts Create Unemployment

- Workers displaced from shrinking sectors do not automatically match into expanding ones.
- Skills may not transfer perfectly.
- Geography matters if jobs grow in different provinces or cities.

Real Example

A decline in manufacturing may reduce jobs in one region while healthcare and technology jobs expand elsewhere. Workers may need retraining or relocation to re-enter employment.

Figure 17.3: Regional Unemployment Rates, 1976–2019



Reading Figure 17.3

- Regional unemployment rates in Canada differ persistently.
- Some provinces face chronically higher unemployment than others.
- Regional shocks do not disappear instantly because moving is costly and local industry structures differ.

Interpretation

Regional unemployment differences suggest that labour markets are not perfectly integrated. Geography is an important part of structural unemployment.

Displaced Workers: Stylized Findings

- Permanent job loss often causes large earnings losses.
- Re-employment frequently occurs at substantially lower wages.
- Losses can persist for many years.
- Older and less-educated workers often have lower re-employment probabilities.

Definition

Job displacement refers to involuntary job loss due to reasons such as plant closure, downsizing, or restructuring.

Consequences of Job Displacement

- Lower future wages
- Longer unemployment spells
- Loss of firm-specific human capital
- Stress on family well-being and mental health

Real Example

A worker laid off after 20 years in a unionized manufacturing plant may eventually find work, but at a lower wage and with weaker benefits in a different sector.

Three Approaches

- **Implicit contracts:** risk sharing, rigid wages, layoffs in bad states.
- **Efficiency wages:** firms pay above-market wages to raise productivity or reduce turnover.
- **Insider–outsider theory:** incumbents influence bargaining while outsiders have little influence.

Interpretation

All three models help explain why wages may stay above market-clearing levels, generating persistent unemployment.

Implicit Contracts: Risk Sharing

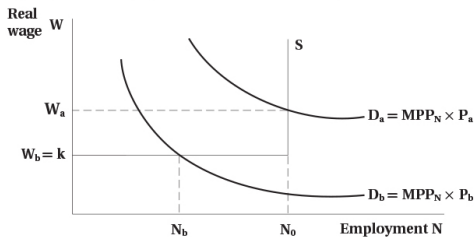
- Workers are often more risk-averse than firms because their human capital is not easily diversified.
- Firms may offer relatively stable wages across states of nature.
- Employment adjusts instead of wages, especially in bad states.

Definition

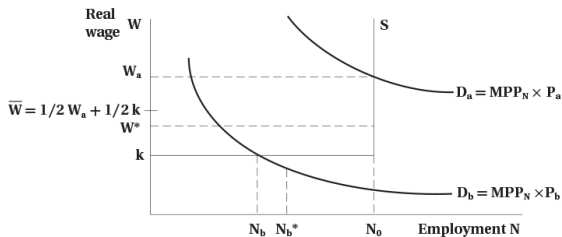
An **implicit contract** is an informal or self-enforcing arrangement where firms smooth wages in exchange for flexibility in employment.

Figure 17.4: Implicit Contracts

(a) Wages and employment with market-clearing



(b) Wages and employment with implicit contracts



Reading Figure 17.4

- In a market-clearing world, wages fluctuate with labour demand.
- Under implicit contracts, wages are smoother.
- Employment absorbs more of the adjustment in bad states.

Interpretation

Risk sharing can make workers better off ex ante, but it can also create layoffs and unemployment when demand is weak.

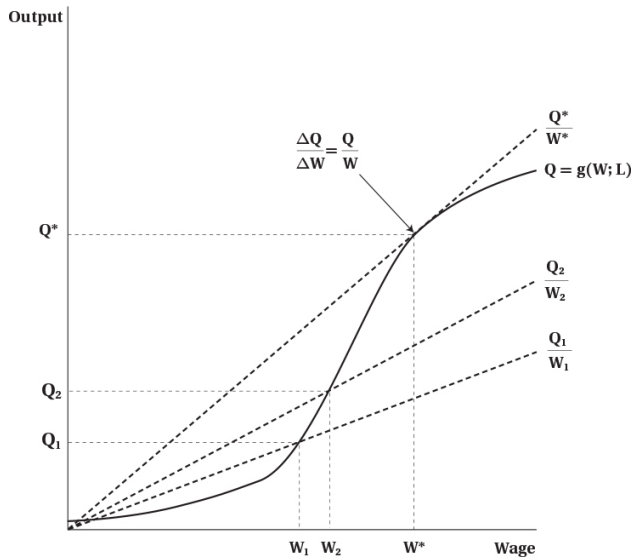
Efficiency Wages: Core Mechanisms

- Higher wages may raise morale and effort.
- Higher wages may reduce shirking and turnover.
- Firms may use high wages to deter unionization or attract better applicants.
- In some models, better nutrition also raises productivity at low wage levels.

Definition

An **efficiency wage** is a wage deliberately set above the market-clearing wage because it improves worker productivity or lowers labour costs through other channels.

Figure 17.5: Determination of the Efficiency Wage



Reading Figure 17.5

- Output rises with the wage because worker efficiency increases.
- But the gain from raising the wage eventually diminishes.
- The optimal efficiency wage is where the gain from better performance is balanced against the higher wage cost.

Interpretation

If firms pay efficiency wages, unemployment can persist because wages stay above the level that would clear the labour market.

- Wages are bargained mainly with incumbent workers.
- Replacing insiders with outsiders is costly because of hiring, training, and disruption costs.
- Excess labour supply from outsiders may not push wages down much.

Definition

Insiders are current employees with bargaining power. **Outsiders** are unemployed or non-employed workers who want jobs but have little influence on wage setting.

Why Insider–Outsider Forces Matter

- Firms may prefer to keep current workers at high wages rather than replace them.
- This can generate wage rigidity and persistent unemployment.
- Outsiders may remain unemployed even if they are willing to work for less.

Real Example

A firm may keep experienced employees at negotiated wages because replacing them would require costly training and risk production disruptions.

- UI/EI provides insurance against income loss from unemployment.
- It also affects incidence and duration of unemployment through incentives and liquidity.

Definition

Unemployment Insurance (UI) or **Employment Insurance (EI)** is a social insurance program that partially replaces lost earnings for eligible unemployed workers.

Why Have UI/EI?

- Workers cannot fully insure themselves against job loss.
- Private insurance markets may fail because job loss risk is correlated across workers during recessions.
- Social insurance smooths consumption and reduces hardship.

Real Example

A worker laid off in a recession may have mortgage or rent payments due immediately. EI helps smooth income while they search for a new job.

Channels: Incidence and Duration

- Higher benefits may increase separations by making unemployment less costly.
- For unemployed workers, higher benefits lower search costs and may lengthen optimal search.
- Empirically, exit rates often rise sharply as benefits approach exhaustion.

Interpretation

UI/EI creates a trade-off: it provides valuable insurance but may also increase unemployment by affecting job separation and job search behaviour.

Canada's UI/EI System: Key Features

- Established in 1940.
- Major reforms in 1971–72.
- Renamed Employment Insurance in 1996.
- Replacement rate around 55% for beneficiaries.
- Claimant types include frequent, long-tenured, and occasional users.

Interpretation

The details of eligibility and benefit duration matter greatly for how EI affects both insurance value and labour market incentives.

Behavioural Effects and Design Considerations

- Moral hazard vs. liquidity effects
- Repeat use and seasonal patterns
- Regional extended benefits
- Labour force participation effects

Definitions

- **Moral hazard:** insurance changes behaviour because the insured event becomes less costly.
- **Liquidity effect:** benefits allow workers to avoid accepting poor job matches immediately.

Interpreting the EI Design Trade-Off

- A well-designed system should protect workers from hardship.
- But it should also preserve incentives for efficient job search and employment.
- The optimal design balances social insurance against induced unemployment.

Real Example

If EI is too stingy, workers may be forced into poor matches quickly. If it is too generous for too long, some may delay search excessively. Good design tries to avoid both extremes.

- Wage rigidity and imperfect information explain how unemployment can coexist with vacancies.
- Search theory uses reservation wages and hazard rates to explain unemployment duration.
- Sectoral shifts and job displacement can generate large and persistent costs.
- Above-market wages can arise through implicit contracts, efficiency wages, and insider–outsider mechanisms.
- UI/EI provides valuable insurance but also affects incidence and duration of unemployment.

Thank You!